

Superstore Sales Performance Analysis Report

Portfolio Project

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Role: Data Analyst

Executive Summary

This report analyses the sales performance of a Superstore using Python and the Superstore dataset. The objective is to identify key sales trends, evaluate business performance, determine profitability drivers, and recommend strategies that improve revenue and profit.

The project follows the complete analytics lifecycle:

- *Data Loading
- *Data Cleaning
- *Exploratory Data Analysis (EDA)
- *Statistical Analysis
- *Business Performance Evaluation
- *Financial Impact Assessment
- *Business Recommendations

The analysis was performed using Python, Pandas, NumPy, Matplotlib and Seaborn.

Project Objectives

The objectives of this project were to:

- *Examine sales performance
- *Evaluate company profitability
- *Identify data quality issues
- *Understand customer purchasing patterns
- *Measure discount impact on profit
- *Provide management recommendations supported by data

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

path = '/content/'
df = pd.read_csv(path + dataset)
df.head()
```

Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
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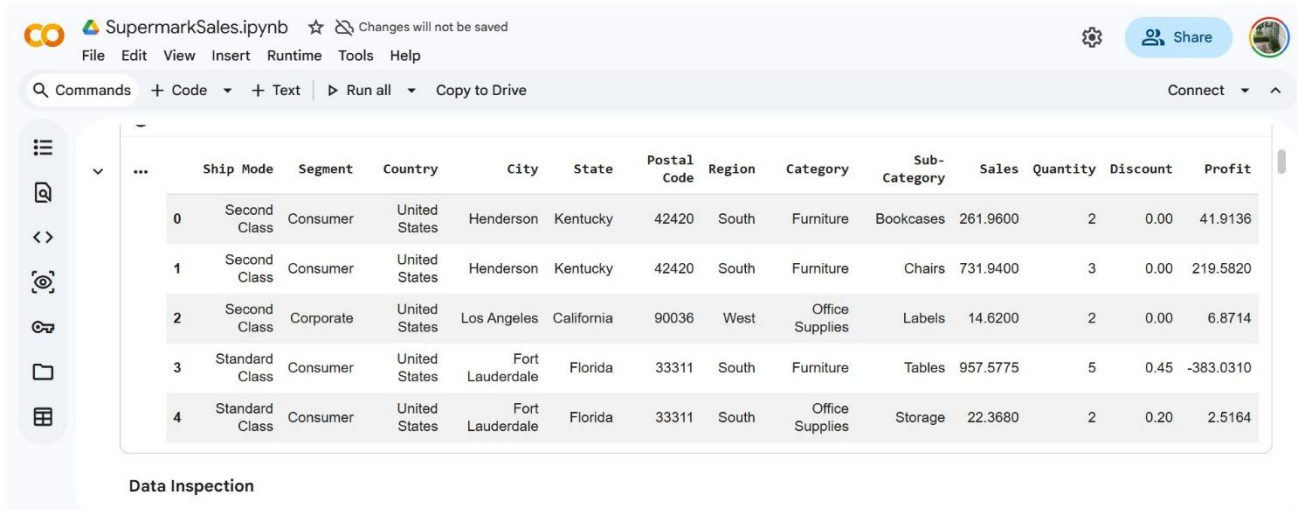
Tools Used

- *Python
- *Google Colab
- *Pandas

- *NumPy
- *Matplotlib
- *Seaborn

Dataset Overview

The dataset contains transactional sales records from a retail superstore.



The screenshot shows a Jupyter Notebook interface with a file named 'SupermarkSales.ipynb'. The top bar includes a search bar, file management icons, and a 'Share' button. The main content area displays a table with 15 columns: Ship Mode, Segment, Country, City, State, Postal Code, Region, Category, Sub-Category, Sales, Quantity, Discount, and Profit. The table contains 5 rows of data. Below the table, the text 'Data Inspection' is visible.

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.9600	2	0.00	41.9136
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.9400	3	0.00	219.5820
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.6200	2	0.00	6.8714
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.0310
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	22.3680	2	0.20	2.5164

Dataset Size

Original Records: 9,994

Variables: 13

Columns include:

- *Ship Mode
- *Segment
- *Country
- *City
- *State
- *Postal Code
- *Region
- *Category
- *Sub-Category
- *Sales
- *Quantity
- *Discount
- *Profit

Data Cleaning

Data quality assessment was conducted before analysis.

Duplicate Records

The dataset contained:

17 duplicate records

Python Code:

```
df.duplicated().sum()
```

Output
17

Duplicates were removed.
`df.drop_duplicates(inplace=True)`

Missing Values
The dataset contained no missing values.
`df.isnull().sum()`

Output
0 Missing Values

This indicates high-quality data that required minimal preprocessing.

Data Inspection

`df.shape`

Output
(9994,13)

After cleaning
9977 records

Exploratory Data Analysis

Descriptive statistics were generated.
`df.describe()`

Key observations
Average Sales:
230.15

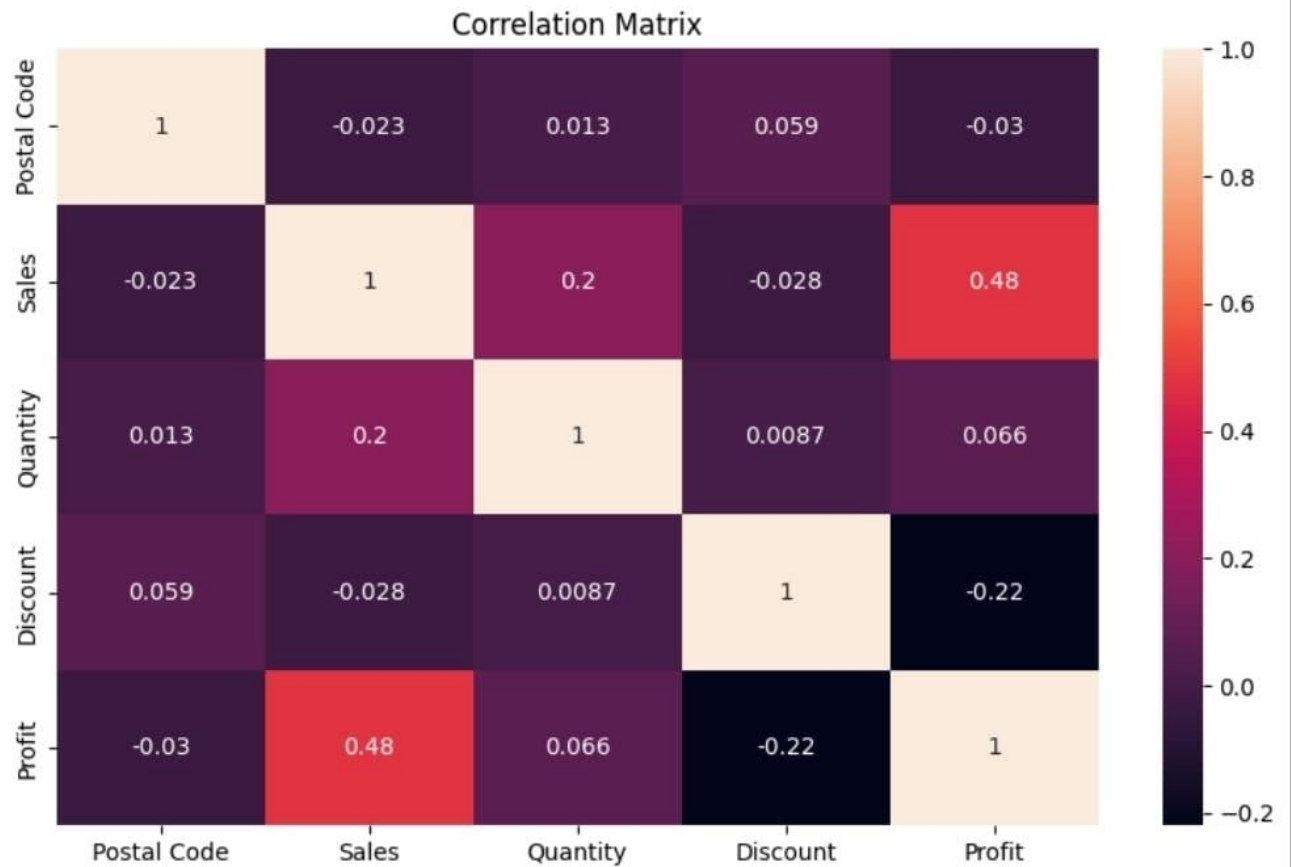
Median Sales
54.82

The large difference between the mean and median indicates the presence of high-value sales transactions.

Correlation Analysis

A correlation matrix was created.

`sns.heatmap(df.corr(numeric_only=True),annot=True)`



Key Findings

Sales and Profit

Positive correlation:
0.48

Interpretation:

As sales increase, profit generally increases.

Discount and Profit

Negative correlation:
-0.22

Interpretation

Higher discounts reduce profitability.

This is an important business insight because discounting is directly reducing company earnings.

Outlier Detection

A boxplot revealed several outliers.

These outliers represent:

- *High-value customers
- *Bulk purchases
- *Exceptional profit transactions

Instead of removing them, they should be investigated because they may represent valuable customers.

Sales Performance

Python Code

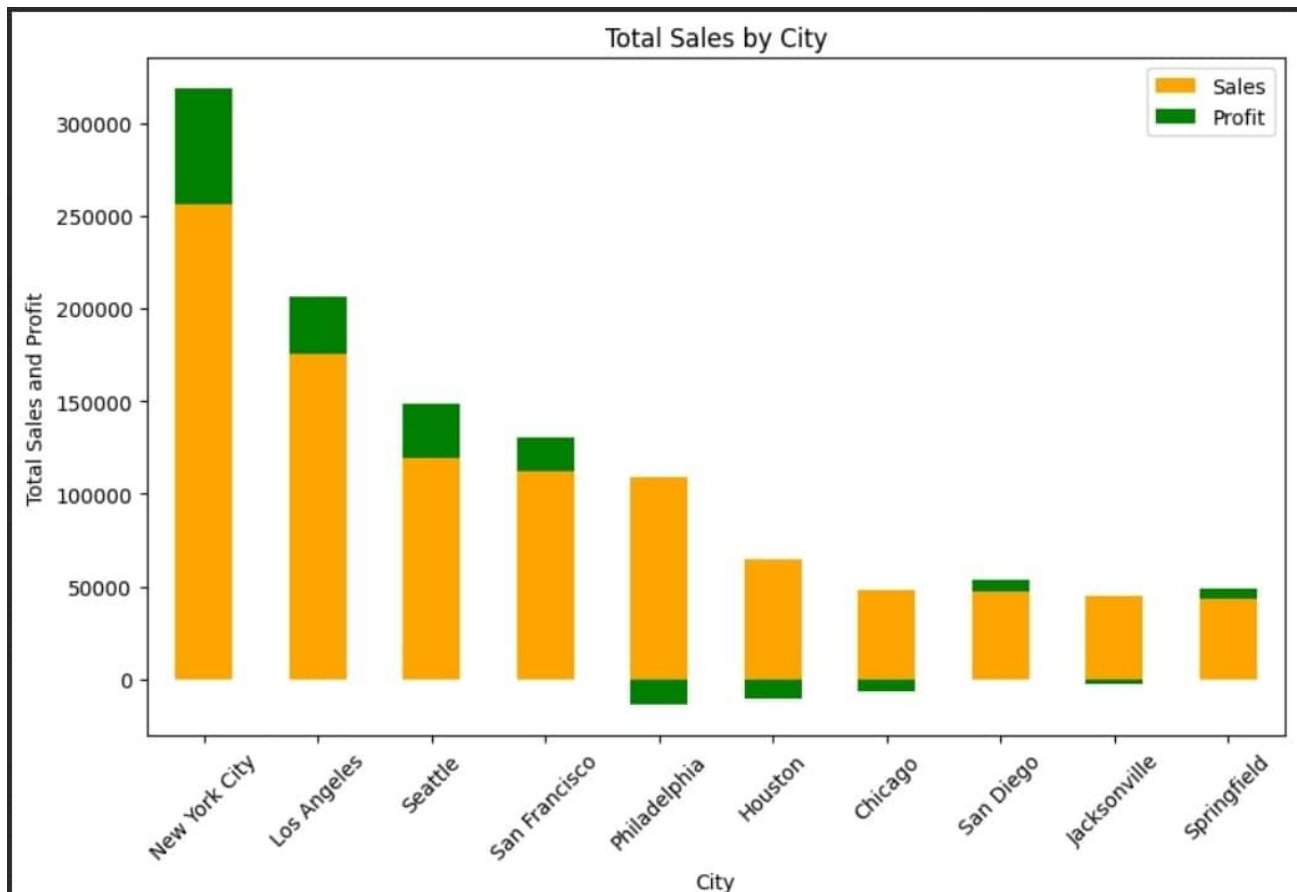
```
total_sales = df['Sales'].sum()  
print(total_sales)
```

Output

Total Sales

\$2,296,195.59

This indicates the company generated approximately \$2.30 million in revenue during the reporting period.



Quantity Sold

Python

```
total_quantity = df['Quantity'].sum()
```

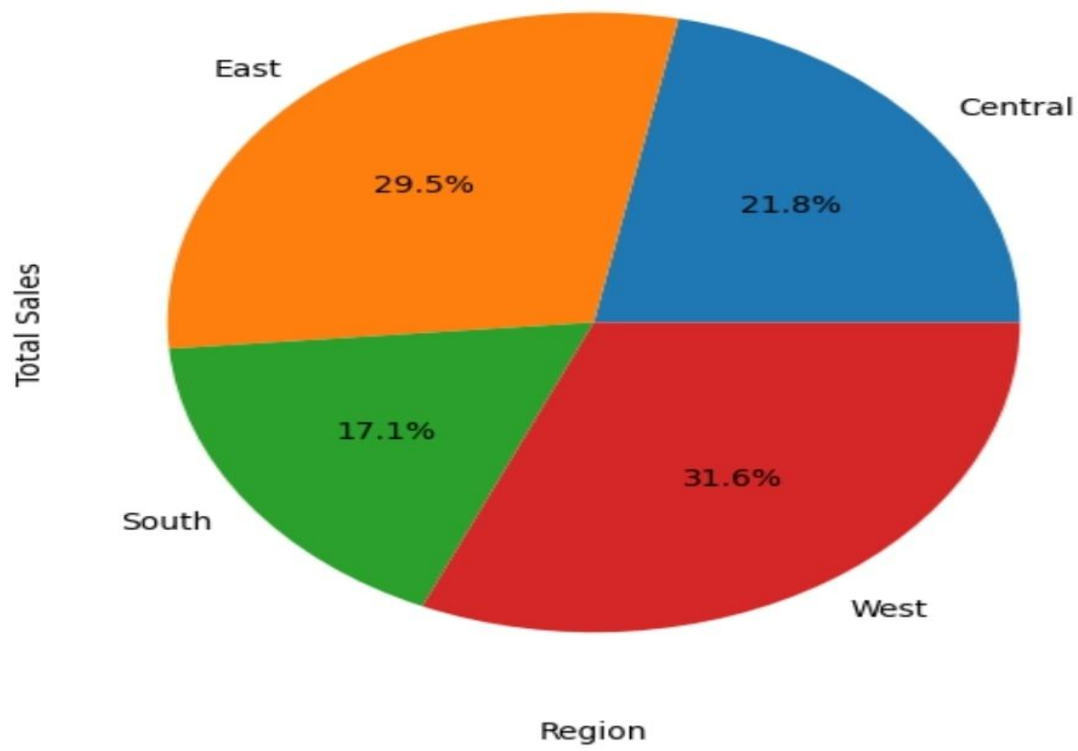
Output

Total Quantity Sold

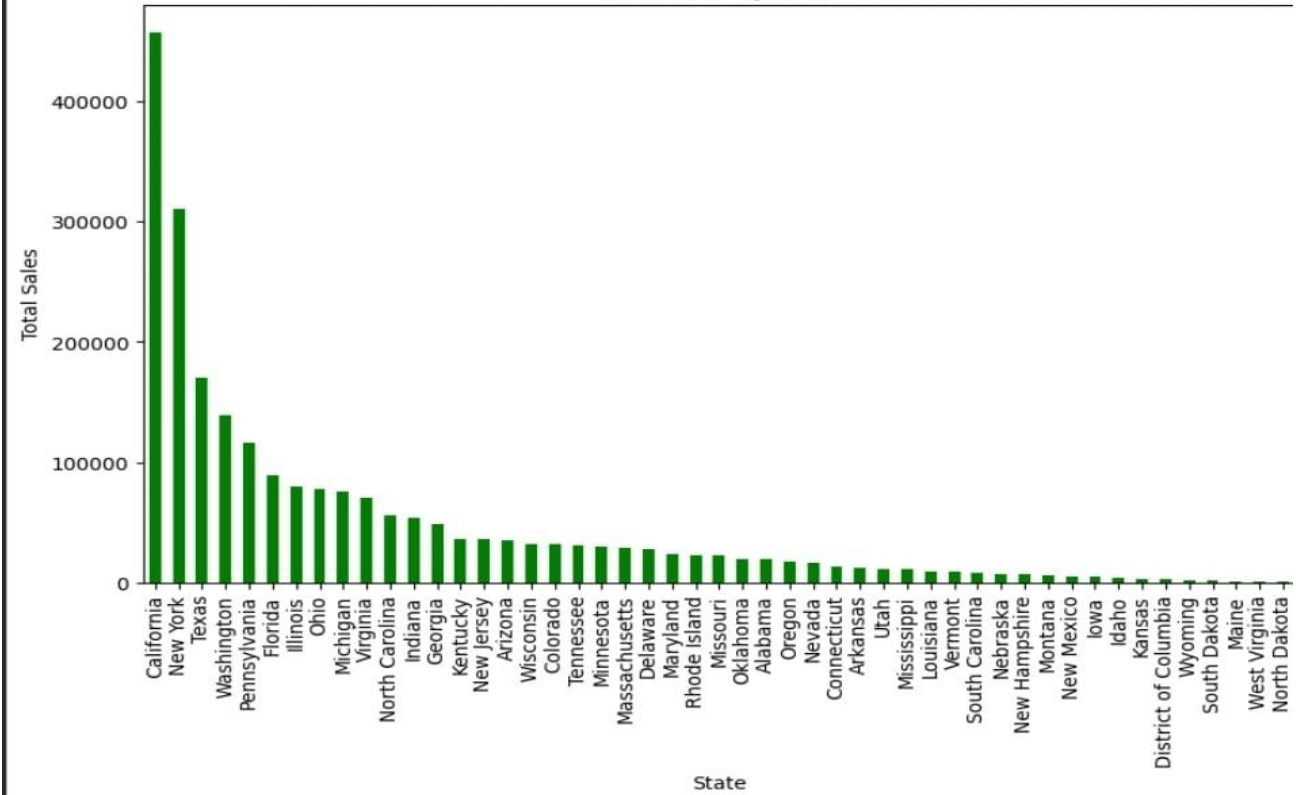
37,820 units

This demonstrates healthy product movement and consistent customer demand.

Total Sales by Region



Total Sales by State



Total Discount

Output

Approximately

1,559

The company frequently uses discounts to stimulate demand.

While discounts may increase sales volume, excessive discounting negatively affects profitability.

Total Profit

Python

```
total_profit = df['Profit'].sum()
```

Output

Approximately

\$286,397

Profit Margin

Profit Margin

$\text{Profit} \div \text{Sales} \times 100$

Calculation

$286,397 \div 2,296,195 = 12.5\%$ Profit Margin

Interpretation

A profit margin of approximately 12.5% suggests that the business is profitable. However, there is significant room for improvement through better pricing, cost control, and discount management.

Business Insights

1. Strong Revenue Performance

The company generated over \$2.29 Million in sales. This demonstrates strong market demand

2. Moderate Profitability

Although revenue is high, profits represent only about 12.5% of total sales.

This indicates that operating costs, discounts, or product margins are reducing profitability.

3. Discounts Reduce Profit

The correlation analysis clearly shows that:

Higher discounts lead to lower profits.

Management should carefully monitor discount policies.

4. Sales Growth Does Not Always Equal Profit Growth

Increasing sales alone does not guarantee increased profit.

Selling low-margin products or offering excessive discounts can increase revenue while reducing earnings.

5. Presence of Premium Customers

Large sales outliers suggest that the company serves high-value customers.

Management should identify these customers and implement loyalty programmes to retain them.

Financial Statement Impact

Income Statement

The sales performance directly affects:

Revenue
↑ Higher Sales
↓
Gross Profit
↓
Operating Profit
↓
Net Profit

Since the company generated over **\$2.29 Million** in revenue, the Income Statement would show strong sales performance. However, high discounts reduce Gross Profit and Net Profit.

Balance Sheet Impact

Sales performance influences several Balance Sheet accounts.

Cash

Higher sales generate higher cash inflows if customers pay immediately. Cash Assets increase.

Accounts Receivable

If customers purchase on credit, Accounts Receivable increases. The company must ensure customers pay on time.

Inventory

As more products are sold,
Inventory decreases.
The company will need to replenish stock regularly.

Retained Earnings

Profit of approximately \$286,397 increases retained earnings. Higher retained earnings strengthen shareholders' equity and improve the company's financial position.

Shareholders' Equity

Since the company is profitable, Shareholders' Equity increases through retained profits.

Cash Flow Impact

Positive sales improve operating cash flow.
However: Excessive discounting reduces cash generated from each sale. Therefore, The company should focus on increasing profitable sales rather than simply increasing sales volume.

Risks Identified

The analysis identified several business risks.

Heavy Discounting

This reduces profitability.
Recommendation: Introduce targeted promotional campaigns instead of blanket discounts.

Profit Dependence on High Sales Volume

The company appears dependent on large sales volumes.
Management should improve product margins to increase profitability.

Customer Concentration

Large outlier transactions suggest dependence on high-value customers. If these customers leave, Revenue could decline significantly.

Strategic Recommendations

Improve Discount Strategy

Only offer discounts where they increase long-term customer value.

Increase High-Margin Product Sales

Promote products with higher profitability instead of focusing only on sales volume.

Customer Segmentation

Segment customers into:

- *Premium customers
- *Corporate customers
- *Regular customers

Develop personalised marketing campaigns for each segment.

Sales Dashboard

Develop an interactive dashboard using: Power BI
to monitor:

- *Sales
 - *Profit
 - *Discount
 - *Inventory
 - *Regional Performance
- in real time.

Predictive Analytics

Use Machine Learning to forecast:

- *Future Sales
- *Inventory Requirements
- *Customer Demand
- *Profit Trends

Sample Python Code Used

Loading Dataset

```
import pandas as pd  
df = pd.read_csv('SampleSuperstore.csv')
```

Check Duplicates

```
df.duplicated().sum()
```

Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

Check Missing Values

```
df.isnull().sum()
```

Descriptive Statistics

```
df.describe()
```

Correlation Matrix

```
plt.figure(figsize=(10,6))
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Matrix")
plt.show()
```

Total Sales

```
df['Sales'].sum()
```

Total Profit

```
df['Profit'].sum()
```

Conclusion

This analysis demonstrates that the Superstore has a strong sales engine, generating over \$2.29 million in revenue with an estimated \$286,397 in profit. The business is profitable, but profitability is constrained by discounting practices and potentially low-margin product sales.

From a financial perspective, sustained profitability will strengthen the company's income statement through higher net earnings and improve the balance sheet by increasing cash (where sales are collected promptly), reducing inventory through product turnover, and growing retained earnings and shareholders' equity. However, if excessive discounts continue, the business risks eroding gross margins and weakening long-term financial performance despite healthy revenue growth.